The Project and Earned Value

Bruce Link and Eleanor Haupt (ASC/FMCE)

Objectives

- Theory behind project requirements
- Basic Earned Value concepts
- Earned Value Management System (EVMS)

What We'll Cover...

- Overview of Comp 4911 project
- Basic EVMS terms
- Earned value and the project management cycle
  - Planning
  - Executing
  - Controlling
- References (for further reading)

Overview of Comp 4911Project

- Build a realistic data entry and project management system for a software development company (the *System*)
- Primary role is project manager
  - Plan and manage software projects
    - budget, work breakdown, assign staff to work packages, compare actual results to planed, using Earned Value
- All employees enter weekly hours worked in time sheets
- Additional roles with the system
  - Human resources assign employees to line managers
  - Supervisors assign their employees to projects
    - Project managers manage employees assigned to their project
  - Timesheet approvers validate timesheets

System Requirements 1/2

- Automate certain project management aspects of software development at a medium size company (1000 software developers)
- Support software development *projects* by capturing information on the *planned* and *actual* effort for each *work package*
- Projects are identified by a unique numeric project code
- All work on a project is organized hierarchically into work packages, each with an id that shows its position in the hierarchy
  - the project will include functionality to manage work package start/end times, project (and engineer) planned effort
- All work in the company is done within some project
- Employees are identified by an employee number, and are assigned to one of more projects
- Employees fill in weekly timesheets, indicating the number of hours charged to each project/work package, for each day of the week

System Requirements 2/2

- The system should accept weekly timesheet entries from the employees (preferably with electronic signature) and enter the data into a database.
- Every week, for each active work package, the engineer responsible for the work package will enter updates to the effort required to finish the work package
- Every month, the system should provide a report comparing the actual and planned effort for each work package on the project. The system must provide Earned Value reports as specified by the customer.
- Project functionality should be able to be entered from any computer that has a web browser (using any operating system)
- The system will use component-based technology and be deployed on OKD

Overview of Earned Value

Why use EVMS?

Life without EVMS

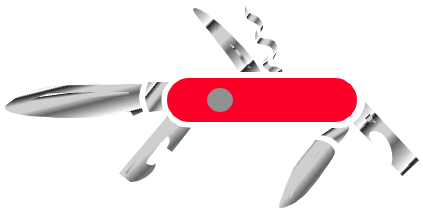
- Given:
  - total budget of $100,000
  - 12 month effort
  - produce 20 units
- Status:
  - spent to date: $64,000
  - time elapsed: 6 months
  - units produced: 8 complete, 2 partial
- **How are you doing, and how do you know how you are doing?**
- **How far along are you? (64%, 50%, >40%)**

Why use EVMS?

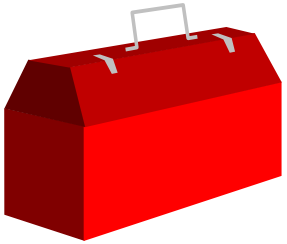
- Early and accurate identification of trends and problems
- Accurate picture of contract status
  - cost, schedule, and technical
- Basis for course correction
- Supports mutual goals of contractor and customer
  - bring project in on schedule and cost

What is EVMS?

EVMS is the primary project management tool...



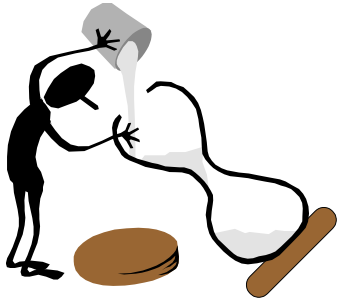
that integrates the technical, schedule, and cost parameters of the contract.



The project manager is the primary tool in the EVMS toolbox.

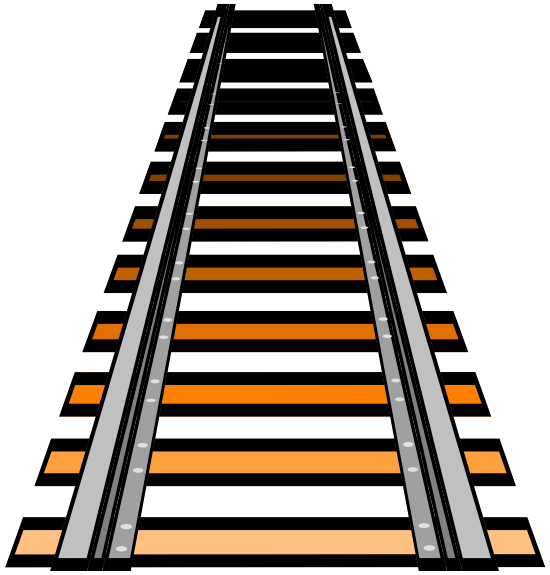
What is the process?

- The contractor establishes a management control system
- An integrated baseline plan is established
  - work is defined, scheduled, and resources are allocated
- Work and resources are driven down to lowest level for execution
- A work authorization system is set up that controls changes to the baseline
- Budgets are “earned” as work is completed = EARNED VALUE
- Status provided against baseline
  - schedule and cost variances are isolated
- Early insight provided into final estimated cost



Basic EVMS Terms

EVMS measures progress



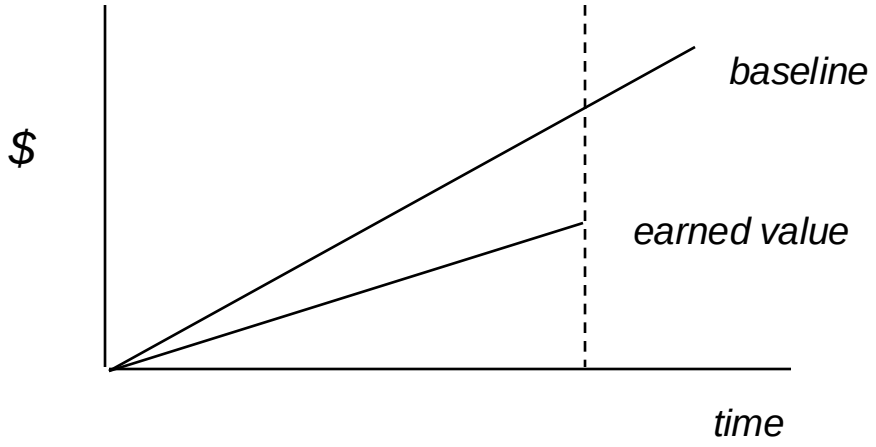
Progress = Movement Forward

to measure progress, there must be a standard against which the forward movement may be compared

EVMS establishes a baseline to measure progress

What do we measure progress against?

- Performance measurement baseline
  - budget that is spread over . . .
  - time, to accomplish the scope of
  - work
  - and against which progress can be measured
- Earned Value is key concept
  - how much progress did I make against my original plan?
  - expressed in dollars or hours



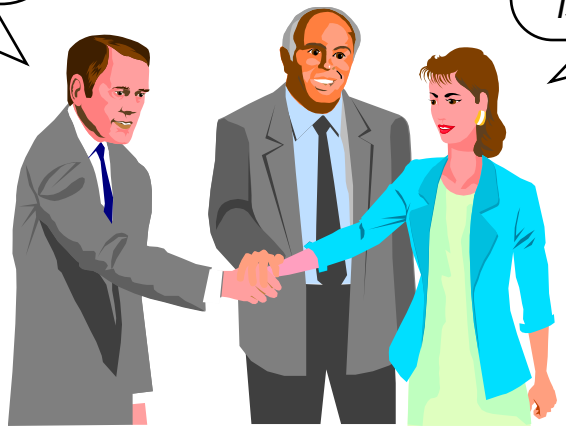
Five Basic Elements

BCWS	Budgeted Cost of Work Scheduled
BCWP	Budgeted Cost of Work Performed
ACWP	Actual Cost of Work Performed
BAC	Budget at Completion
EAC	Estimate at Completion

Total Budget

It's my pleasure to award you this contract for a new railroad track

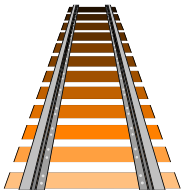
hmm...5 miles of track, 5 months to do it all....$5000 budget....This is going to be tough!



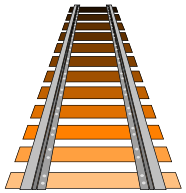
What is the total job supposed to cost?

What is the value of the contract at cost?

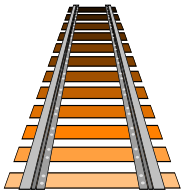
Budgeted Cost of Work Scheduled (BCWS)



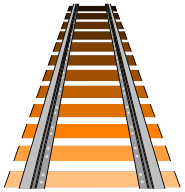
Month 1
BCWS = $1,000



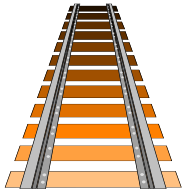
Month 2
BCWS = $1,000



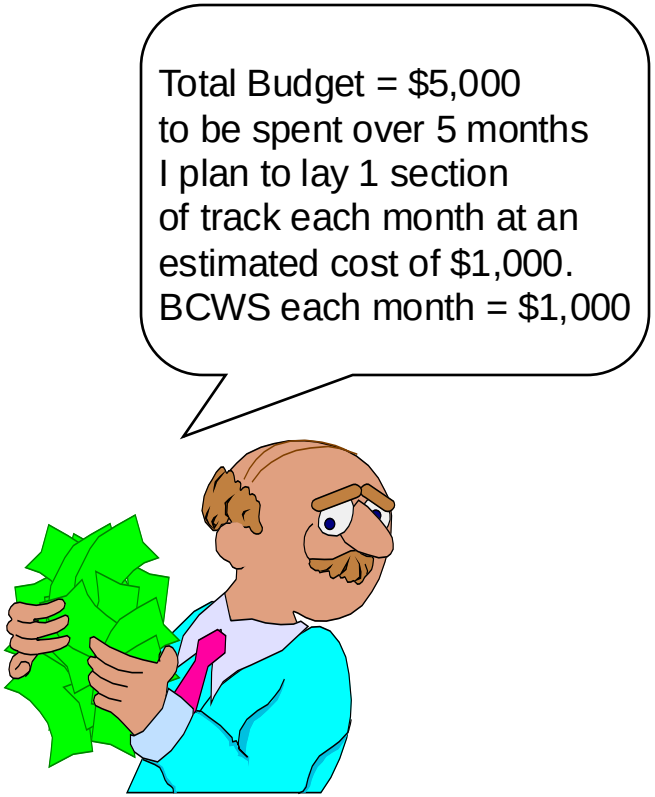
Month 3
BCWS = $1,000



Month 4
BCWS = $1,000

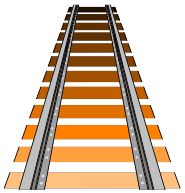


Month 5
BCWS = $1,000

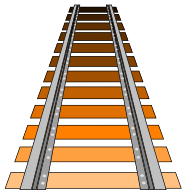


each dollar of BCWS represents a specific dollar of work scope

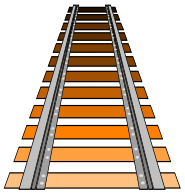
Budgeted Cost of Work Scheduled (BCWS)



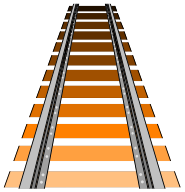
Month 1
BCWS = $1,000



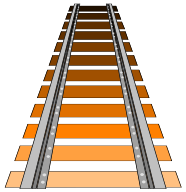
Month 2
BCWS = $1,000



Month 3
BCWS = $1,000



Month 4
BCWS = $1,000



Month 5
BCWS = $1,000

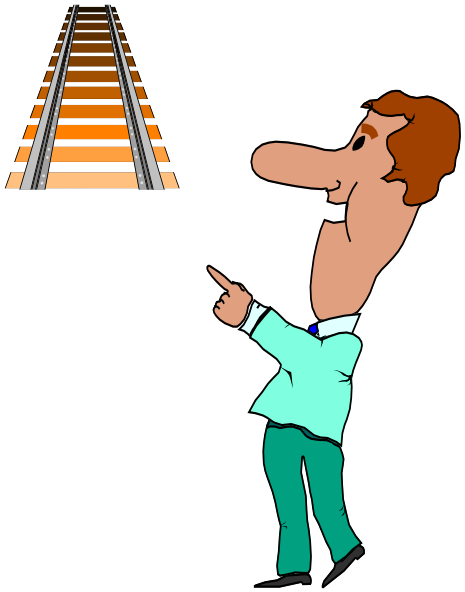
Total Budget = $5,000

Total BCWS = $5,000

BCWS is aggregated and summed as the performance measurement baseline

Budgeted Cost of Work Performed (BCWP)

the EARNED VALUE concept



We’re at the end of the second month, but only 1 section of track is complete. Value of work performed = $1,000

You earn value the same way as it was budgeted in baseline

Schedule Variance

BUDGET BASED

BCWS — of the work I scheduled to have done, how much did I budget for it to cost?

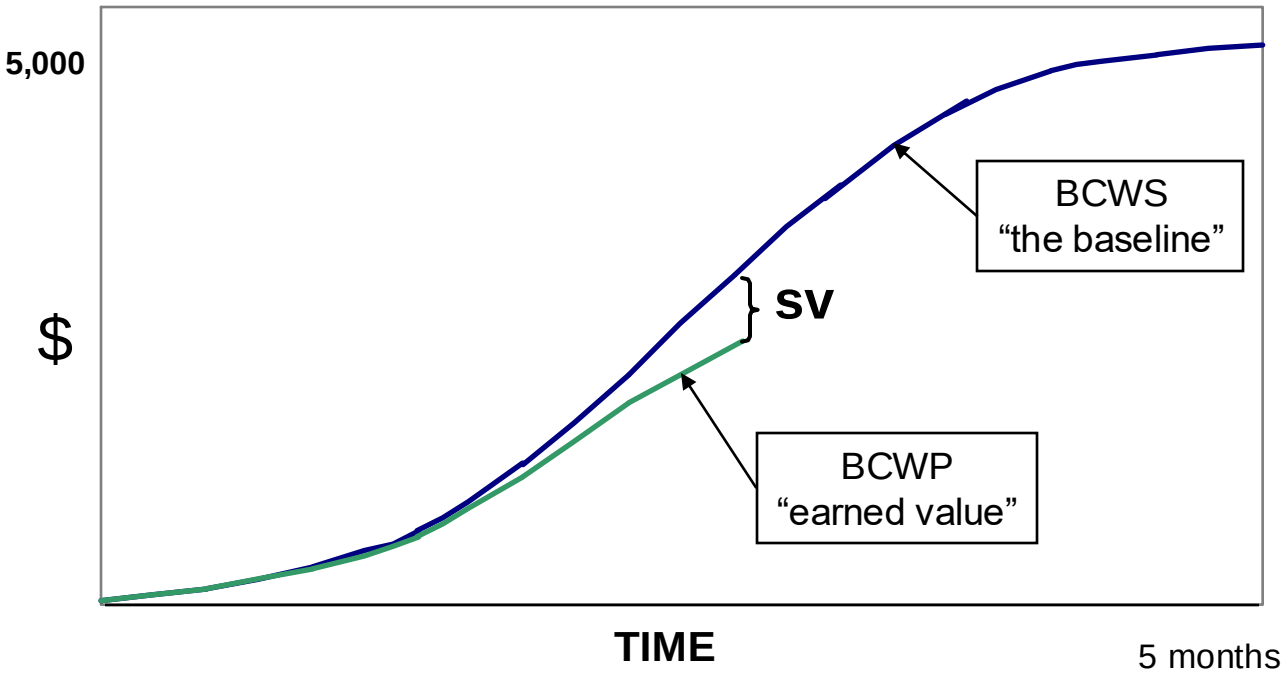
BCWP — of the work I actually performed, how much did I budget for it to cost?

SCHEDULE VARIANCE is the difference between work scheduled and work performed (expressed in terms of budget dollars)

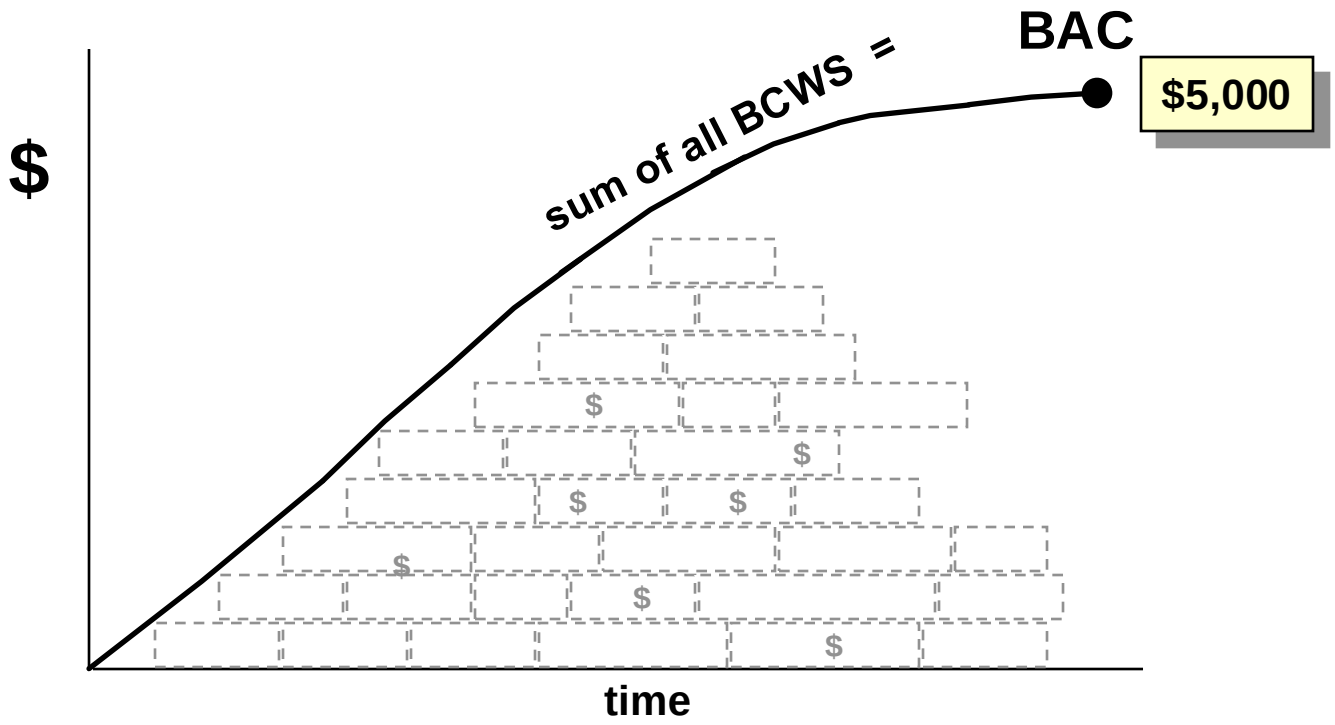
formula: **SV $ = BCWP - BCWS**

example: SV = BCWP - BCWS = $1,000 - $2,000
SV= -$1,000 (negative = behind schedule)

Schedule Variance



Budget at Completion (BAC)

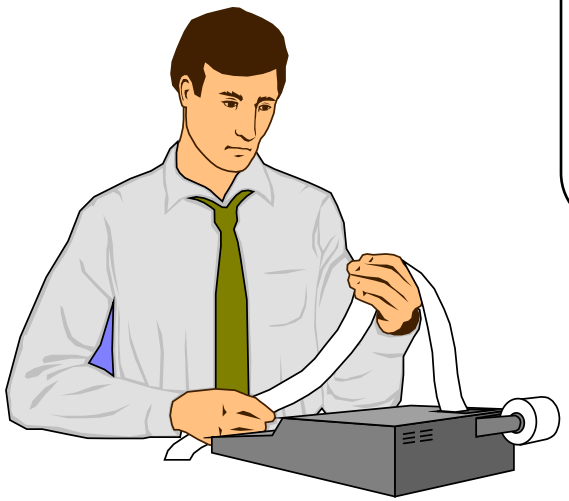


- **when all work has been phased, cumulative BCWS = BAC**
  e.g., $5,000 = $5,000

At the end...

- At the **end** of the contract, when all work has been completed:
  - I've "earned" all of my budget ($5,000)
    - BCWP (cumulative) = **$5,000**
    - BCWS (cumulative) = **$5,000**
    - therefore, schedule variance ($) = 0
  - Formal schedule will reflect whether milestones were achieved on time
- Example:
    - I finished late, but I did finish
      - SV ($) = $ 0
      - Formal schedule shows a 5 month actual delay in completing the contract

Actual Cost of Work Performed (ACWP)



Labor came to $1,300, and materials cost $1,100. That first section of track cost $2,400!

actual expenditures vs. budget

Converting Hours to Dollars

- Budgets are managed in dollars, but people think in hours or days
- To convert, need pay rates
  - how much is one hour by one person worth?
  - how much is a project charged for each hour an employee works
- Usually, companies establish fixed labour rates and assign each employee to one labour rate
  - Project uses P1, P2, P3, P4, P5, JS, SS as labels for the labour rates
  - For each fiscal year there is a value for each rate
    - $ per person-day
    - How much project is charged for people in that rate category

Cost Variance

BC WP

AC WP

PERFORMANCE BASED

of the work I actually performed,
how much did I budget for it to cost?

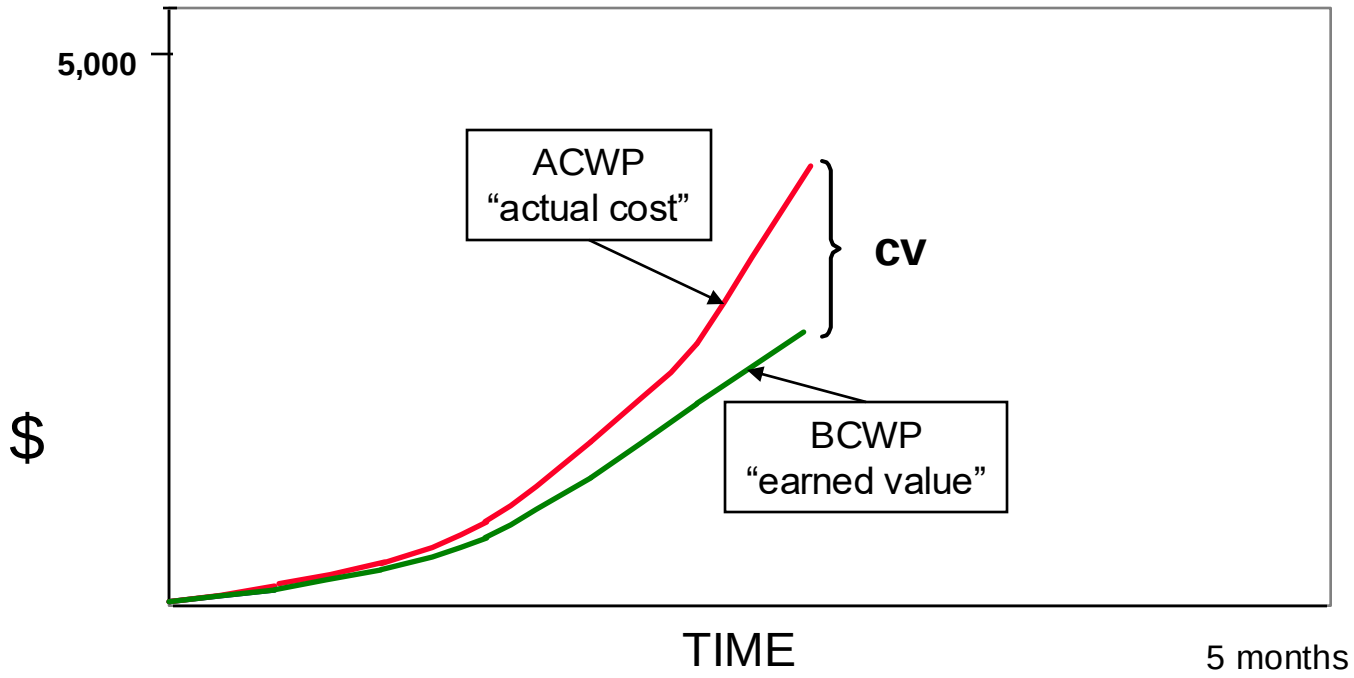
of the work I actually performed,
how much did it actually cost?

COST VARIANCE is the difference between budgeted cost and actual cost

formula: **CV $ = BCWP - ACWP**

example: CV = BCWP - ACWP = $1,000 - $2,400
CV= -$1,400 (negative = cost overrun)

Cost Variance



Estimate at Completion (EAC)

Variance at Completion (VAC)

B AC — what the **total** job is <u>supposed</u> to cost

E AC — what the **total** job is <u>expected</u> to cost

VARIANCE AT COMPLETION is the difference between what the total job is supposed to cost and what the total job is now expected to cost.

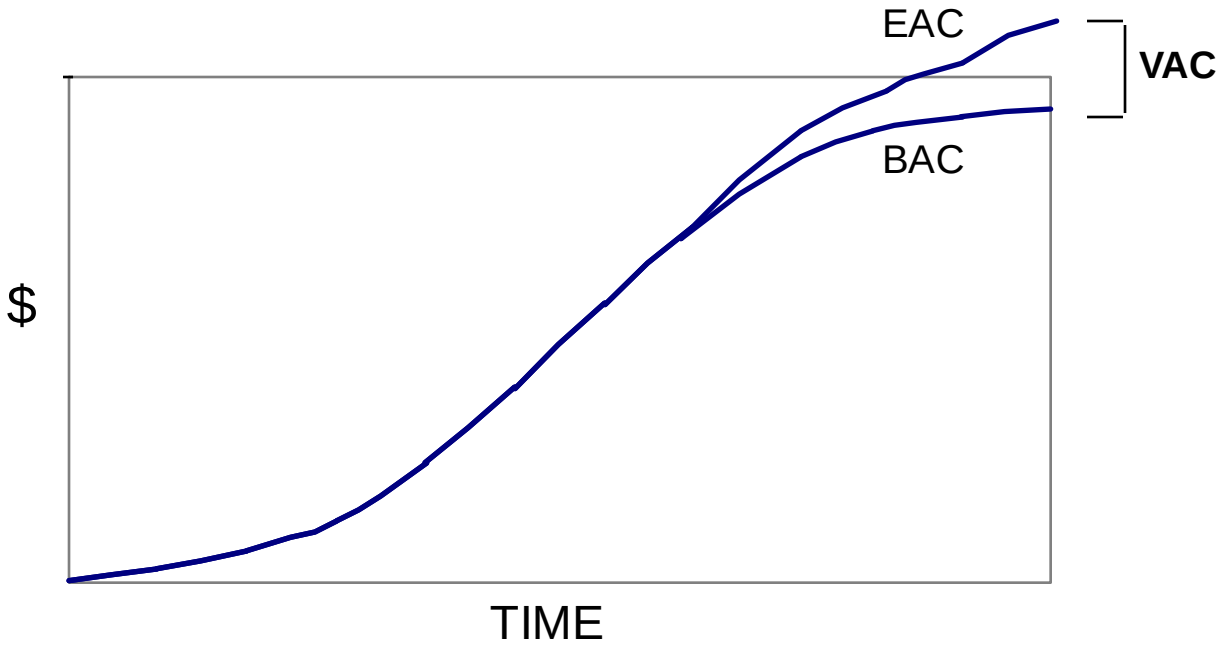
FORMULA: **VAC = BAC - EAC**

Example: VAC = $5,000 - $7,500
VAC = - $2,500 (negative = overrun)

Variance at Completion (VAC)

VAC = Budget at Completion - Estimate at Completion

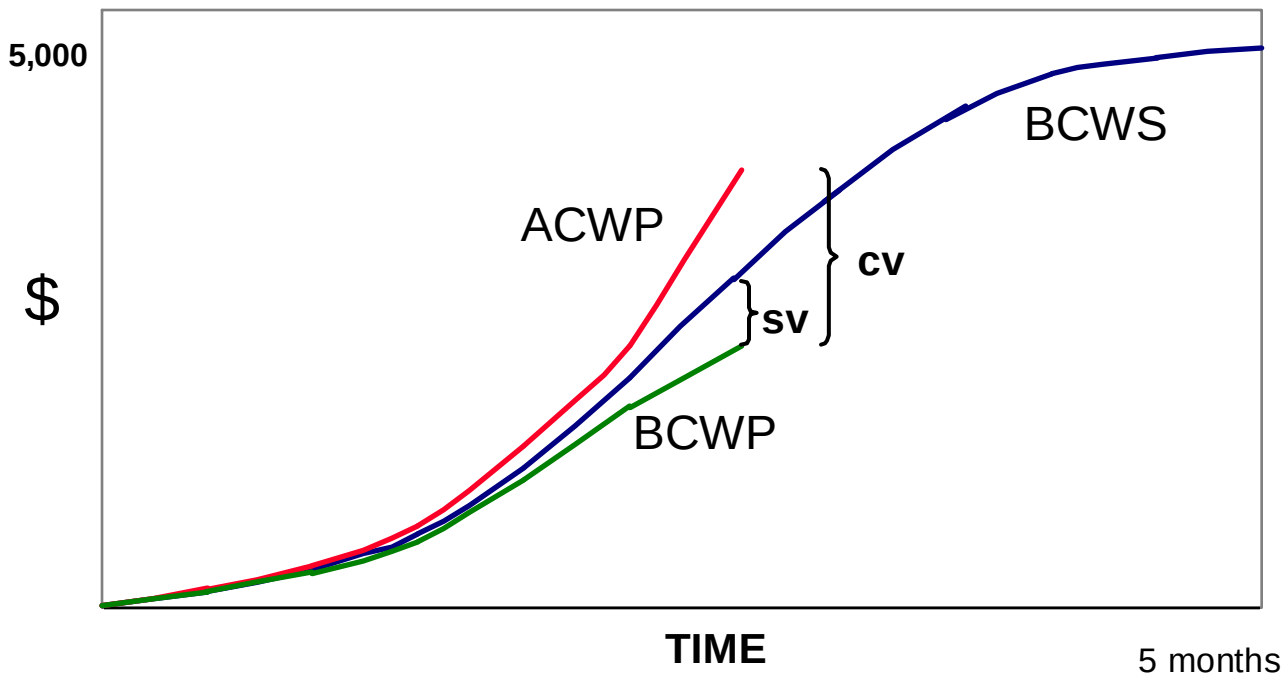
= BAC - EAC



FIVE BASIC PERFORMANCE DATA QUESTIONS & ANSWERS

QUESTION	ANSWER	ACRONYM
How much work should be done?	Budgeted Cost for Work Scheduled	BCWS
How much work is done?	Budgeted Cost for Work Performed	BCWP
How much did the is done work cost?	Actual Cost of Work Performed	ACWP
What was the total job supposed to cost?	Budget at Completion	BAC
What do we now expect the total job to cost?	Estimate at Completion	EAC

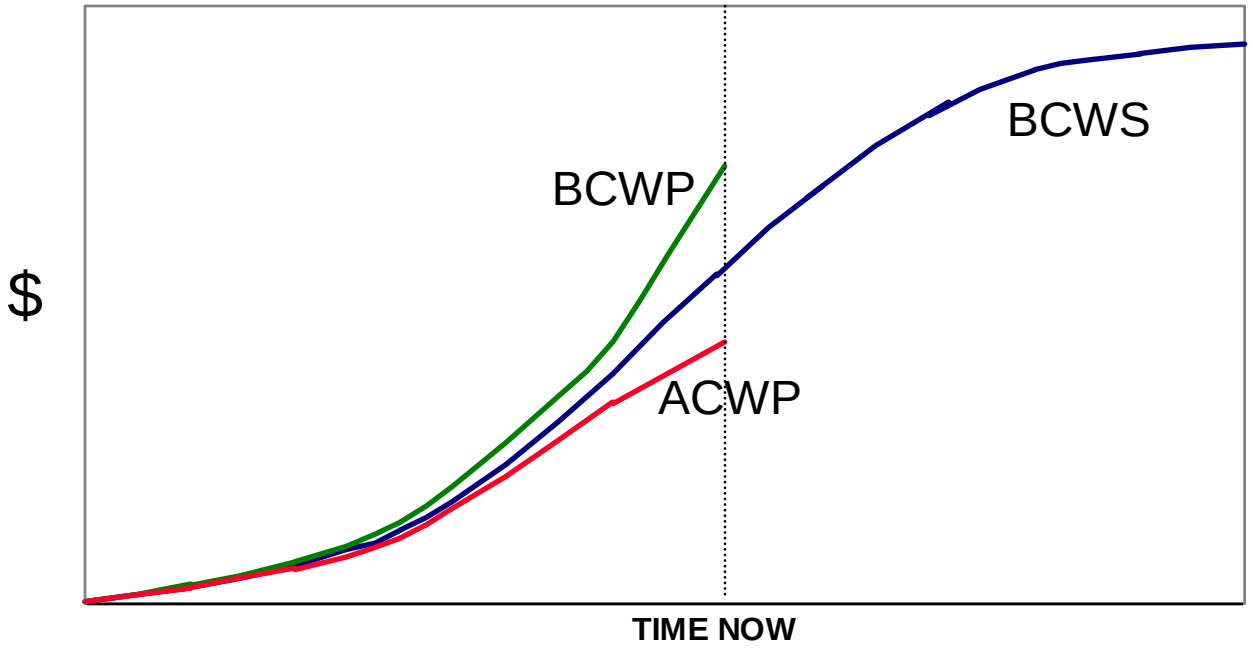
BCWP Allows Isolation of Schedule and Cost Variances



schedule variance = BCWP - BCWS = negative number
cost variance = BCWP - ACWP = negative number

→ **behind schedule, over cost**

Pop Quiz



schedule variance = BCWP - BCWS =
cost variance = BCWP - ACWP =

→ **_____schedule,**
_____cost

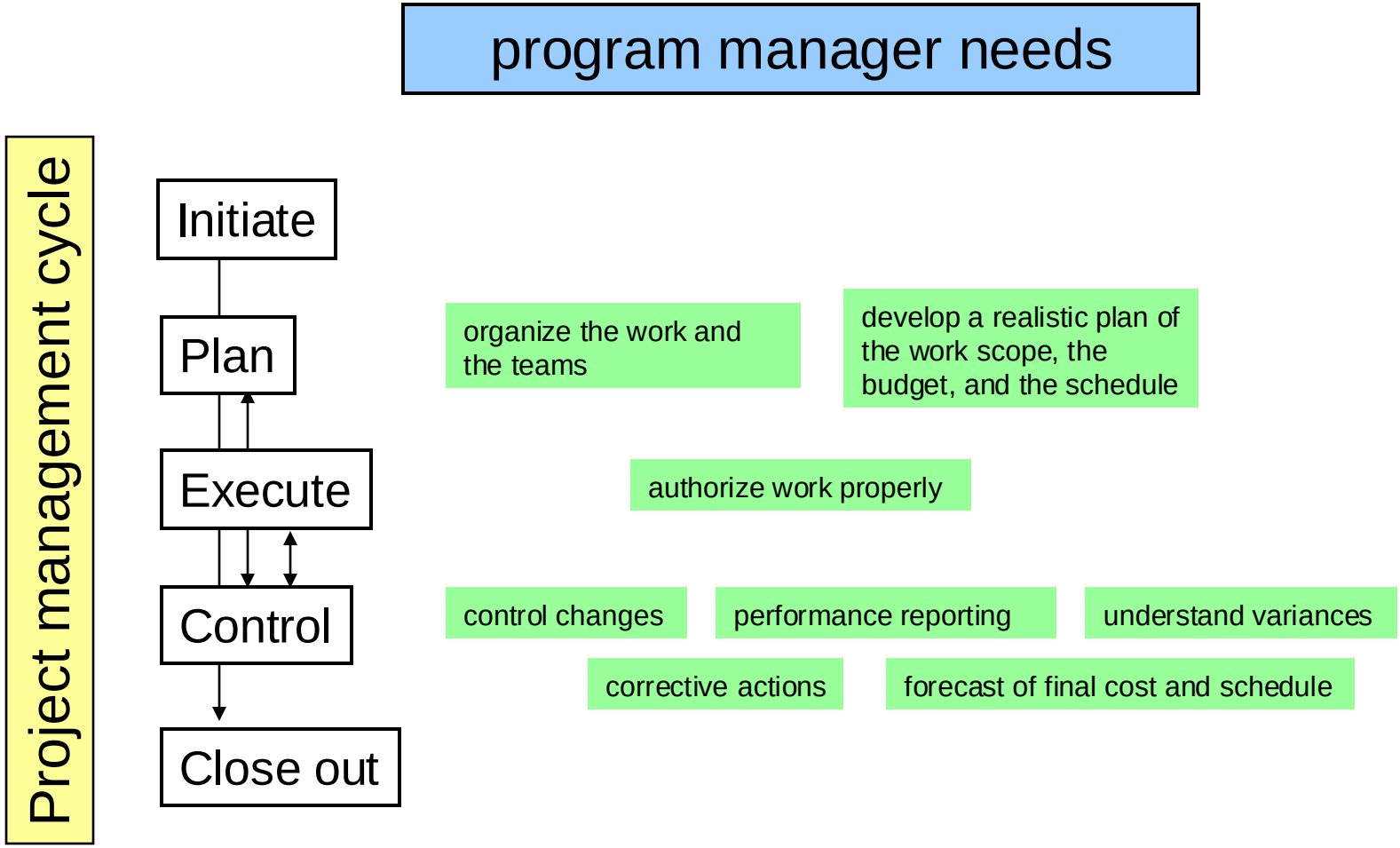
Earned Value Management
and
the Project Management Cycle

Project Management Processes

Initiating	Recognizing that a project or phase should begin and committing to do so
Planning	Devising and maintaining a workable scheme to accomplish the business need that the project was undertaken to address
Executing	Coordinating people and other resources to carry out the plan
Controlling	Ensuring that project objectives are met by monitoring and measuring progress and taking corrective action when necessary
Closing	Formalizing acceptance of the project or phase and bringing it to an orderly end

Source: A Guide to the Project Management Body of Knowledge™, published by Project Management Institute

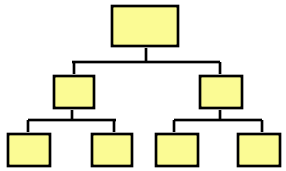
How does EVMS fit into Program Management?



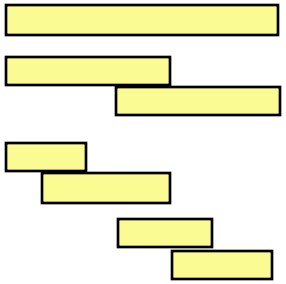
Planning

Planning is a 3 Step Process

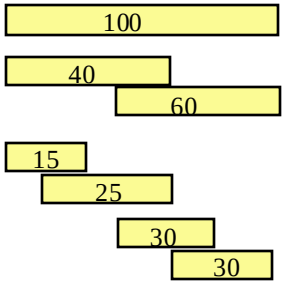
1. DEFINE THE WORK AND ORGANIZE TEAMS

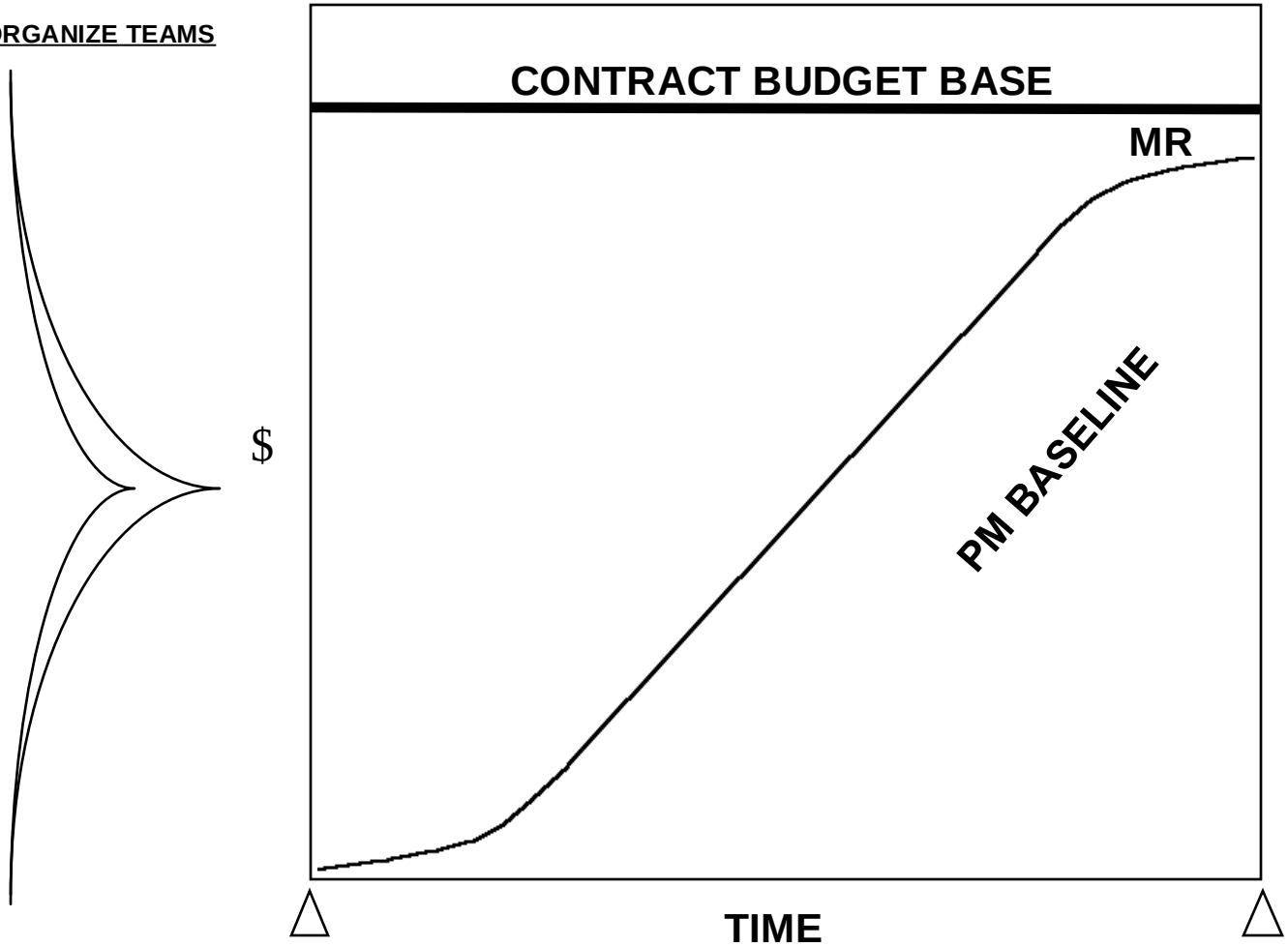


2. SCHEDULE THE WORK



3. ALLOCATE BUDGETS





Organizing

- Objectives of organizing
  - establish a clear picture of the total project work scope
    - do we know what all of the work is?
  - assign responsibility to the right people

The Organizing Process

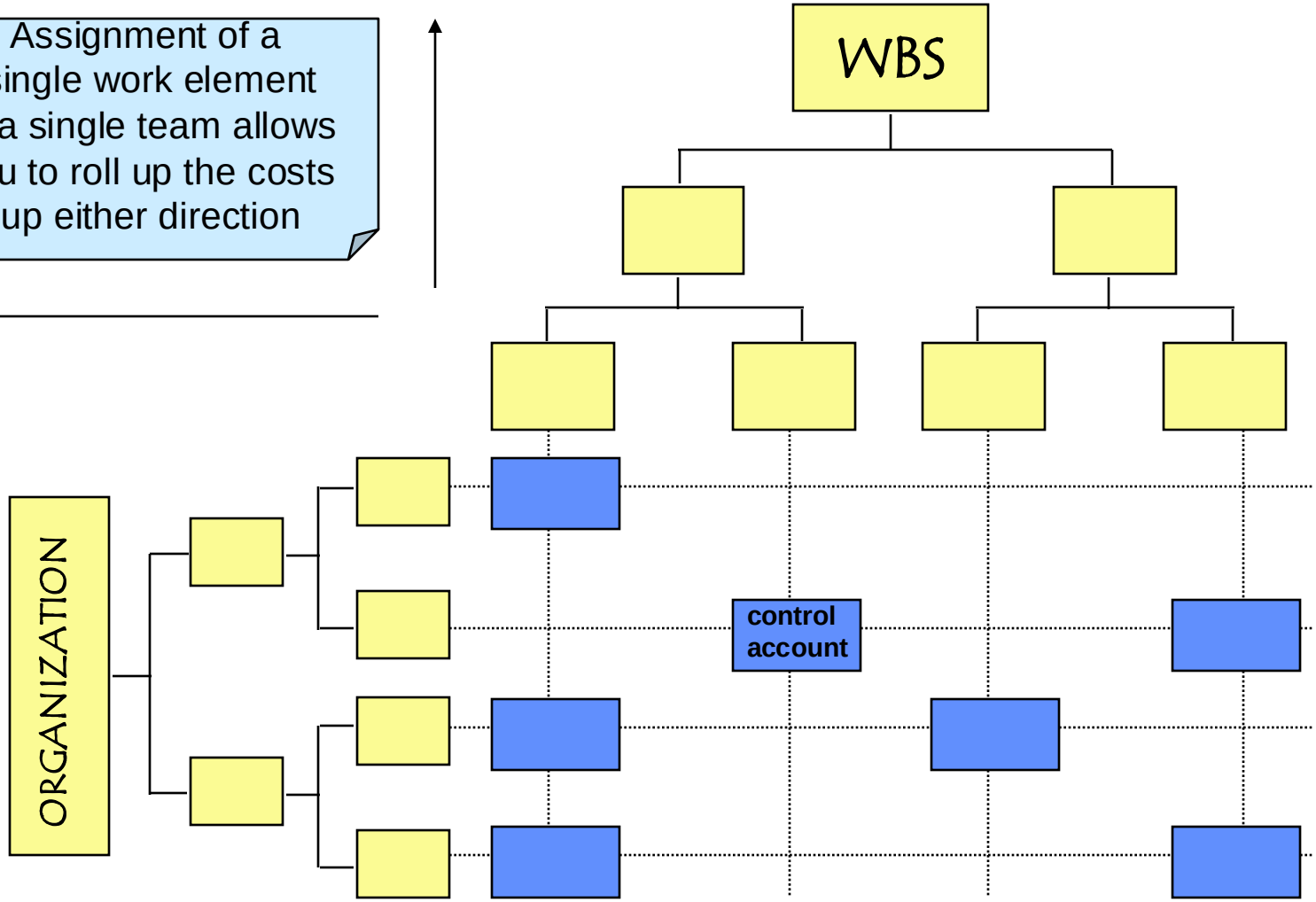
- Process
  - **Step 1**: define the authorized work using a work breakdown structure (WBS)
    - "decompose the work" into manageable chunks
    - provides a framework for
      - program and technical planning
      - cost estimating and resource allocation
      - performance measurements and status reporting
  - **Step 2**: define the organizational structure
  - **Step 3**: assign a single chunk of work to a single manager
    - **control account manager (CAM) or Responsible Engineer (RE)**

Organizing the Work and Teams

- **Customer responsibility**
  - Define upper levels of the WBS (to level 3)
    - MIL-HDBK-881, Work Breakdown Structure
    - Write initial WBS dictionary and include in request for proposal
  - Specify performance reporting levels
- **Contractor responsibility**
  - Extend WBS to level where work is performed
    - define the elements
    - extend WBS dictionary
  - Identify organizational structure
    - include major subcontractors

Assigning Work

Assignment of a single work element to a single team allows you to roll up the costs up either direction



Control Account

- Control Account - where the work is done
  - Intersection of WBS and organization
- Develop Responsibility Assignment Matrix (RAM)
  - contractor developed
  - Assigns work and resources at lowest level (control account)
  - Establishes responsibility for WBS elements
    - Control Account Manager (CAM)
    - include responsibility for overhead

sum of all control accounts (BAC) = complete statement of work

Integration at the lowest level

- **Control Account - the key control point**
  - integrates:
    - work scope
      - WBS element
      - organization
      - work authorization
    - schedule
    - time phased budget (BCWS)
    - actual cost accumulation (ACWP)
  - earned value determination (BCWP)
  - variance analysis (cost and schedule)
    - calculation and explanation
  - corrective action
  - estimate at completion (EAC)

Case Study

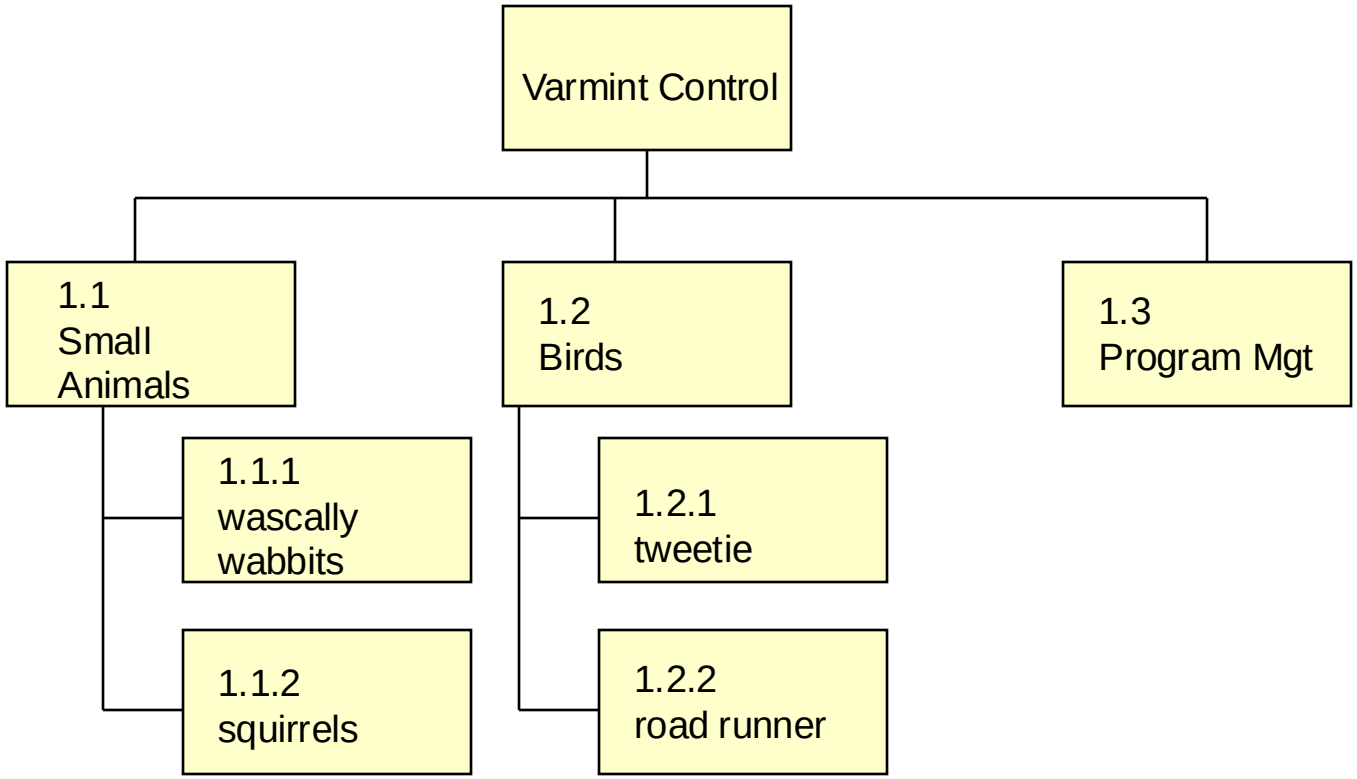
Part 1

Contract Award

- You are the program manager, I. M. Taz
- You just won a contract to eliminate varmints within the state of Arizona
  - birds (tweetie and road runner types)
  - small animals
- You have an organization of highly trained specialists
  - L. M. Fudd
  - Sil Vester the cat
  - Wile E. Coyote
  - Daffie Duck (your deputy and the CAM for management)
- You have allocated the following budgets from your $50,000 award
  - wascally rabbits ($5,000)
  - squirrels ($5,000)
  - tweetie birds ($20,000)
  - road runners ($10,000)
  - program management ($10,000)

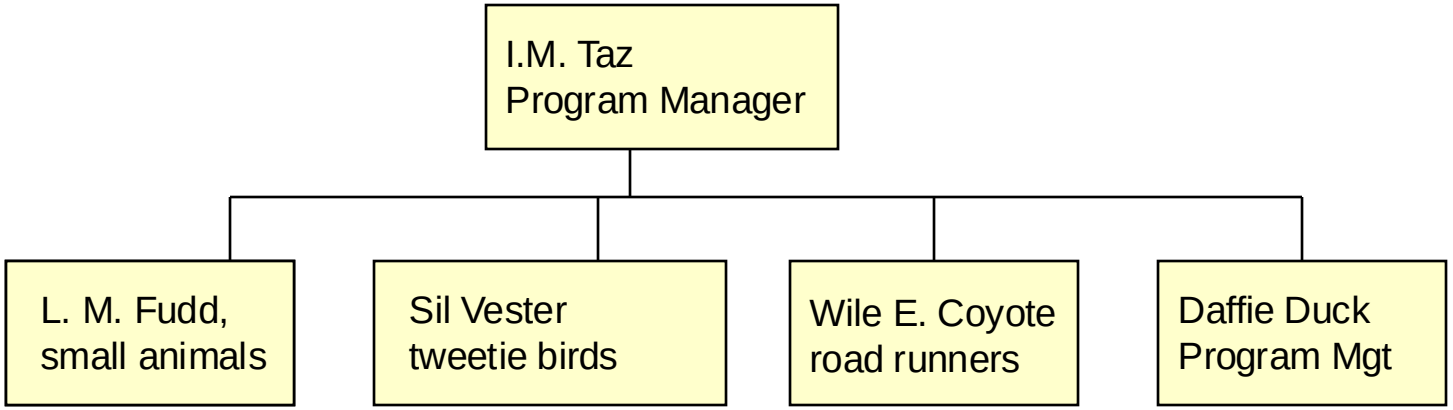
Organize the work

- Build a simple work breakdown structure



Organize the workers

- Build a simple organization breakdown structure



Build a RAM and allocate work

	Sil Vester	L. M. Fudd	Wile E. Coyote	Daffie Duck
1.1.1 wascally wabbits				
1.1.2 squirrels				
1.2.1 tweetie birds				
1.2.2 road runner				
1.3 program management				

Build a RAM and allocate work

	Sil Vester	L. M. Fudd	Wile E. Coyote	Daffie Duck
1.1.1 wascally wabbits		$5,000		
1.1.2 squirrels		$5,000		
1.2.1 tweetie birds	$20,000			
1.2.2 road runner			$10,000	
1.3 program management				$10,000

What makes a good control account?

- **Rules of Thumb for Control Account size**
  - integrated cost and schedule baseline
    - three legged stool
      - legs (cost, schedule, technical) are equal
  - homogeneity of work
  - what is logical to manage every day
  - look at:
    - character of work
    - breakout of labor
    - span of control
  - typically
    - 6 - 18 months for discrete effort
    - level of effort can be longer

What's in a Control Account?

SOW

1.3.4.1
Build ejection seat

SCHEDULE

TIER 1

TIER 2

BUDGET

Labor	1,000 hrs
Labor $	75,000
Material $	25,000

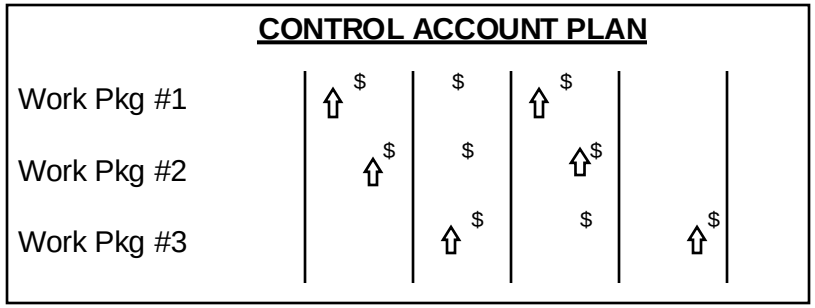
CONTROL ACCOUNT PLAN

CAM name: _____ WBS: _____ Total Budget: _____

Work	$	$	$	
Work	$	$	$	
Work		$	$	$

Work Package

- **Development of Control Account Plans**
  - MAY break down the control account budget into smaller work packages
- **Work Packages**
  - subset of control account
  - reasonably short in duration
  - single element of cost (e.g., labor)
  - single technique for earning value
  - consistent with detail schedules
  - has same characteristics as control account
    - scope of work
    - milestone completion criteria
    - single performing organization
    - start and end dates



Work Package Characteristics

- discrete and measurable
- products or accomplishments
- examples:
  - design drawing package
  - conduct design review
  - install rudder
- rolling wave
  - detailed plans made for near term work packages
  - planning packages are for future work and are not detailed
  - CAMs periodically plan another increment of work packages
- open vs. closed packages

Ways of Earning Value

- **Earned Value techniques**
  - **Discrete**
    - physical, tangible end product
  - **Apportioned**
    - discrete, dependent on another discrete work package
    - example: quality assurance
    - planned as historical estimating factor (e.g., 7%)
  - **Level of Effort**
    - no tangible end product
    - basis of measurement: time
    - when clock starts ticking, you automatically accumulate earned value
    - no schedule variance
    - example: management personnel
- **Should be a quantitative and discrete way to measure the work**
- **May tie in with success criteria or technical measure**
    - e.g., successful completion of a specific test, reliability growth curve

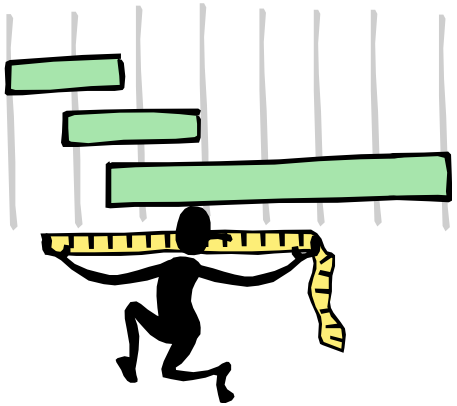
Be Discrete!

- **Discrete EV Techniques:**

Method	How Value is Earned
0/100	no EV at opening, 100% EV at close of WP
50/50	50% EV at opening, 50% EV at close of WP
Units Completed	same budget value for identical units
Equivalent Units	planned unit standards, allows partial credit
Weighted Milestone	each milestone weighted based on planned resources ideal to have a milestone each month
Percent Complete	subjective (least desirable)

Scheduling

- **Scheduling system characteristics**
  - complete
    - all work included
  - formal
    - everyone uses same schedule
  - traceable
    - vertical (Master, Intermediate, Detail)
    - horizontal (between tasks)
  - consistent
  - identifies sequence of tasks, interdependencies



CASE STUDY
Part 2

Contract SOW

SOW Paragraph 1.2.2:

> The contractor shall design, build, and install a system to capture and eliminate the species “Road Runner” within the state of Arizona.

Contractor’s Winning Design:

> The basic system shall consist of five miles of road, a fake tunnel painted on a side of the mountain, plus a device to drop an anvil on the Road Runner.

BEEP BEEP INDUSTRIES

CONTROL ACCOUNT AUTHORIZATION

Control Account: Roadrunner

Control Account Manager: Wile E. Coyote

Reason for Issue: Contract to rid Arizona of all unwanted creatures,
F33657-96-C-0221

Scope Description: Perform scope in accordance with Statement of Work.
WBS Element 1.2.2

Schedule Requirement: Perform in accordance with Control Account Plan

Budget Authorization: $10,000

Schedule Information

- The contract was awarded in Month 1 and will be complete by the end of Month 6.
- You can begin building the road immediately, and plan on it taking about one month to complete one mile of road.
- It will take approximately three months to develop, build, and quality test the anvil support mechanism (ASM).
- You should begin the ASM install during the last month of the build cycle, during quality test. The ASM will take three months to install and should be the last item completed on the contract, in Month 6.
- Based on past experience, you believe that it will take you two months to paint the fake tunnel. You will start it one month before the anvil support mechanism (ASM) begins installation.
- The anvil supplier, Acme Anvils, has been a good supplier for you in the past. The anvil is commercial off-the-shelf equipment. You need delivery one month before the install is complete.

Budget Estimate - BAFO

	Budget Estimate
1. Procure anvil (sole source - ACME Anvil)	$1,500
2. Paint fake tunnel	$1,000
3. Build 5 miles of road	$3,000
4. Develop and build anvil support mechanism (ASM)	$3,000
– design drawings complete & signed off (CDR) (Milestone 1)	($1,000)
– build unit (Milestone 2)	($1,000)
– quality test (Milestone 3)	($1,000)
5. Install system on-site	$1,500
Total	$10,000

EV Techniques	0/100, 50/50, Units Complete, % Complete, Milestones

CONTROL ACCT. TITLE: Roadrunner

CONTROL ACCOUNT MANAGER: Wile E. Coyote

BUDGET: $10,000

	TIER I MILESTONE		⇧ KT AWD					⇧ KT COMP	
WP#	WORK DESCRIPTION	EV METHOD	MONTH 1	MONTH 2	MONTH 3	MONTH 4	MONTH 5	MONTH 6	TOTAL BAC
1	Procure Anvil	**BCWS**							
		BCWP							
2	Paint Fake Tunnel	**BCWS**							
		BCWP							
3	Build Road	**BCWS**							
		BCWP							
4	Build ASM	**BCWS**							
		BCWP							
5	Install ASM	**BCWS**							
		BCWP							
TOTAL CONTROL ACCOUNT PLAN		**BCWS**							
		BCWP							
Schedule Variance		month							
		cumulative							
Actual Costs									
Cost Variance		month							
		cumulative							

EV Techniques	0/100, 50/50, Units Complete, % Complete, Milestones

CONTROL ACCT. TITLE: Roadrunner

CONTROL ACCOUNT MANAGER: Wile E. Coyote

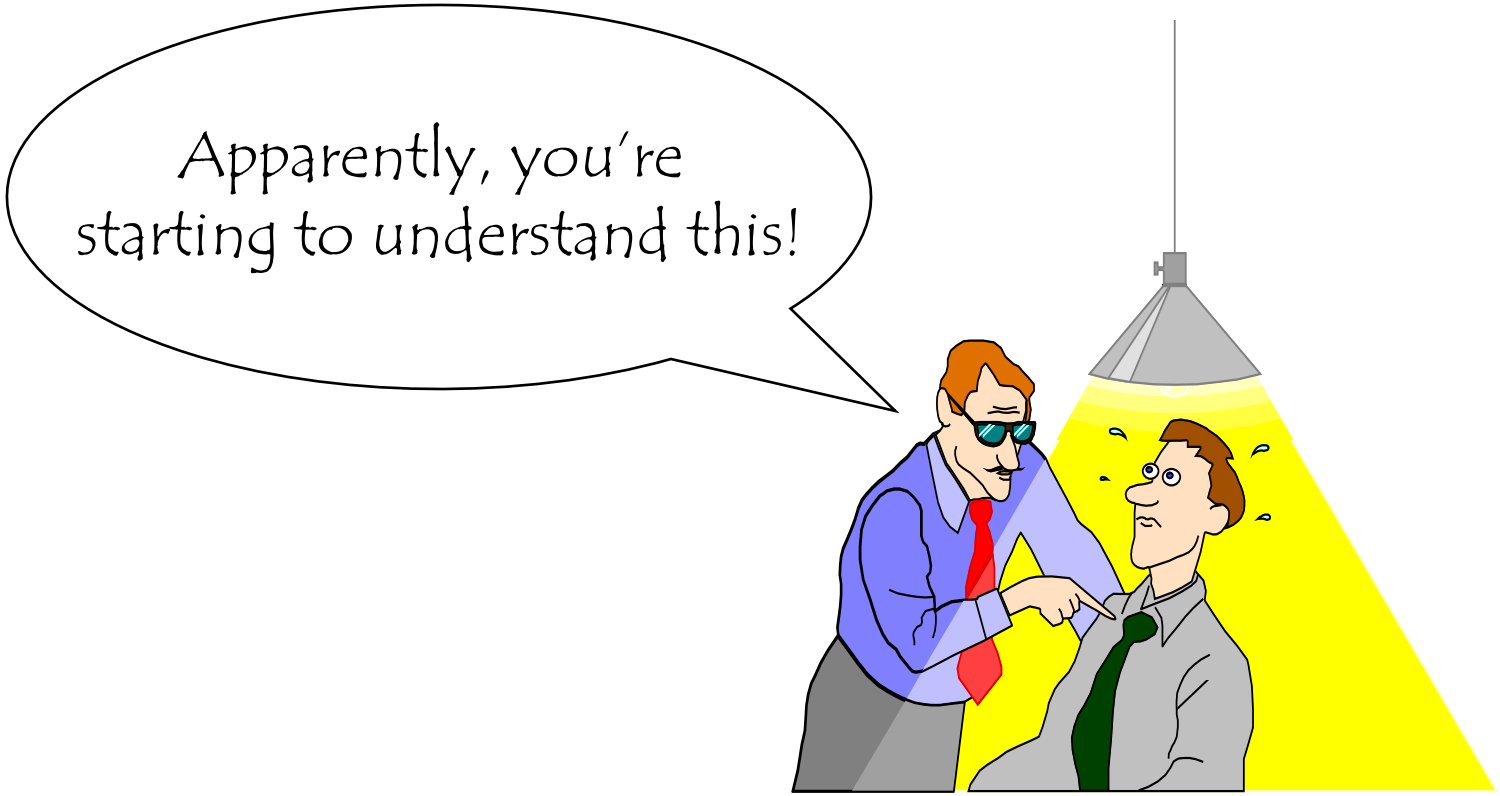
BUDGET: $10,000

	TIER I MILESTONE			KT AWD					KT COMP	
WP#	WORK DESCRIPTION	EV METHOD		MONTH 1	MONTH 2	MONTH 3	MONTH 4	MONTH 5	MONTH 6	TOTAL BAC
1	Procure Anvil	0/100	**BCWS**					1,500		1,500
			BCWP							
2	Paint Fake Tunnel	50/50	**BCWS**			500	500			1,000
			BCWP							
3	Build Road	units complete	**BCWS**	600	600	600	600	600		3,000
			BCWP							
4	Build ASM	milestone	**BCWS**		1,000	1,000	1,000			3,000
			BCWP		1	2	3			
5	Install ASM	% complete	**BCWS**				500	500	500	1,500
			BCWP							
TOTAL CONTROL ACCOUNT PLAN			**BCWS**	600	1,600	2,100	2,600	2,600	500	10,000
			BCWP							
	Schedule Variance	month								
		cumulative								
	Actual Costs									
	Cost Variance	month								
		cumulative								

basic rules of the road...

- **value is earned using the same method as it was planned**
- **sum of the work packages equals the control account budget**
- **sum of the control accounts equals the budget baseline**
- **span of lower tier schedules supports upper schedules**

More Acronyms!



at the total contract level....

UB **Undistributed Budget**

= authorized work held at top level until it can be planned in detail
(will eventually have performance measurement)

PMB **Performance Measurement Baseline**

= time phased budget plan
= detailed planning + UB

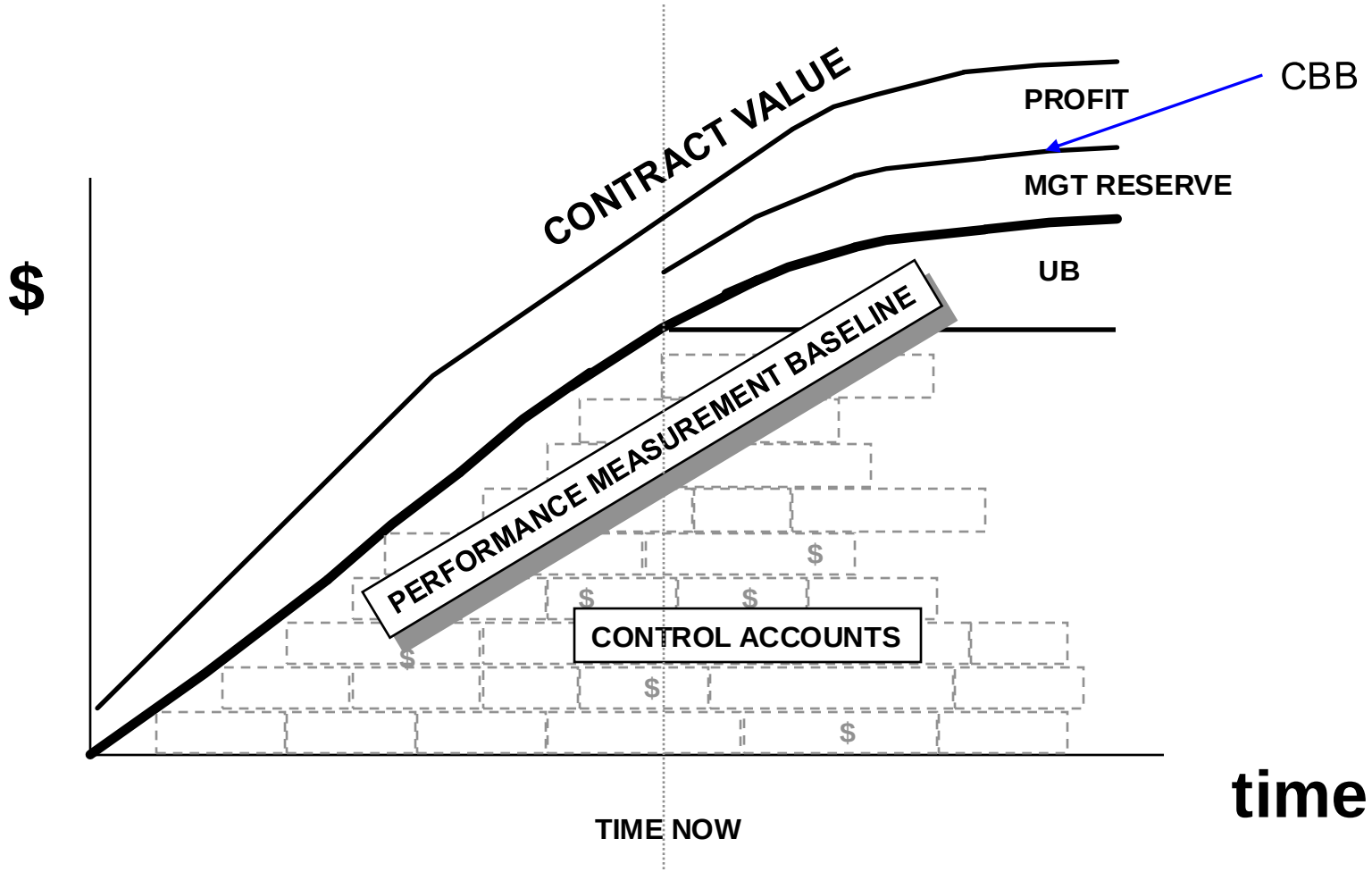
MR **Management Reserve**

= amount withheld at top level for control purposes
(no performance measurement)
= used for unforeseen changes that are <u>within scope</u> of the contract

CBB **Contract Budget Base**

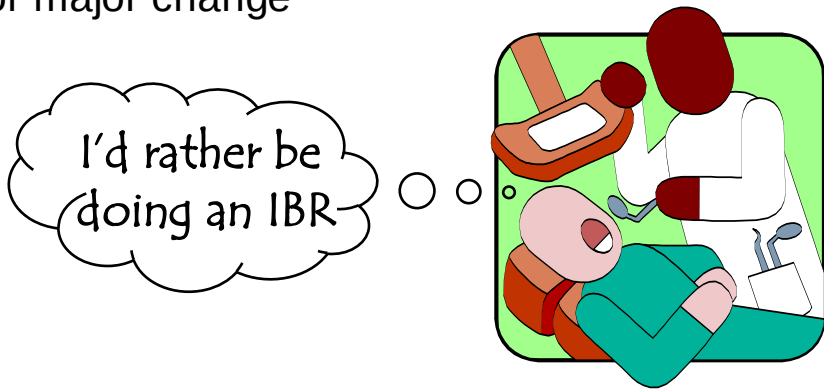
= PMB + MR
= contract at cost

Rolling Up the Work



Plans are in place..

- The baseline is now in place
  - you've broken down all the work
  - assigned it to teams
  - scheduled the work and integrated the schedules,
  - and assigned budget resources…..
  - schedules and budgets roll up to match the contract
- Let's take the time to evaluate the realism of the baseline
  - Integrated Baseline Review (IBR)
  - joint contractor/customer team
  - within 6 months of contract award or major change



Case Study IBR

Historical Data from other Beep Beep programs

1. Average historical cost per mile of road = $750
2. Contractor has no experience in painting tunnels.
3. Contractor has never worked in this part of the country before.
4. Price of raw aluminum on the open market just recently skyrocketed due to heavy demand.

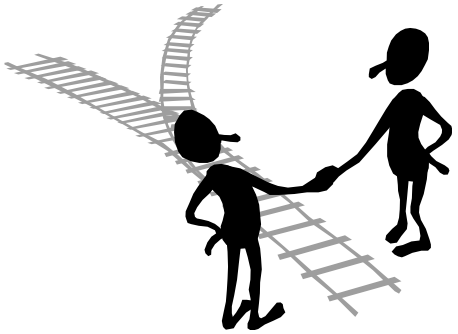
- Did we fully plan all work? Do we understand the work?
- Do we have a reasonable schedule, with logic indicated?
- Do we have enough budget?
- Are the earned value techniques valid?
- Is the program manager paying attention?

- Bottom line: where are the risks to the program?

Execute!

Authorizing the Work

- Can only charge to open work packages
  - contractor system sets procedure
- Contractor maintains baseline log which tracks:
  - distribution of budget from Undistributed Budget (UB) to control accounts
  - distribution of Management Reserve (MR)
  - additions of authorized work
  - total equals Contract Budget Base
- Contract changes incorporated in disciplined manner
  - cannot start work without authorization or budget
- Baseline changes are controlled
  - Internal replanning
  - Over Target Baseline, Over Target Schedule



more rules of the road...

- **cannot move budget and work independently**
- **cannot use management reserve to cover overruns**
- **may replan open work packages as necessary**
  - **contractor sets internal policy**
  - **maintain valid performance information**
- **cannot change budgets or costs for completed work**
  - **except to fix errors**

Case Study
Part 3

EV Update and End of Month Actuals

- Month 1: 1 section of road built ($700)

- Month 2: Design drawings were completed and signed off ($980)
  1 section of road built ($720)

- Month 3: The work package for the fake tunnel was opened ($200)
  A labor strike prevented a section of road from being built ($300)
  The ASM began to be built, but the unit was not complete ($800)

- Month 4: The tunnel was not completed ($400)
  The crew went back on the job, and got paid overtime. 2 road sections built for $1500.
  Milestone 2 for the ASM was finally complete. Quality test was pushed out 1 month. ($400)
  The install of the ASM was delayed, due to the delay in build.
  The anvil was ordered.

- Month 5: The anvil was delivered and final cost was $1,700.
  The tunnel was complete (painted). ($500)
  The last section of road was built for $700.
  Qual test completed. ($1,900)
  The CAM estimated that the install was approximately 20% complete ($400)

- Month 6: Additional work crews were hired, and the installation was completed.
  The additional crews cost an additional $1,000. ($2,100 total)

							EV Techniques	0/100, 50/50, Units Complete, % Complete, Milestones	

CONTROL ACCT. TITLE: Roadrunner CONTROL ACCOUNT MANAGER: Wile E. Coyote

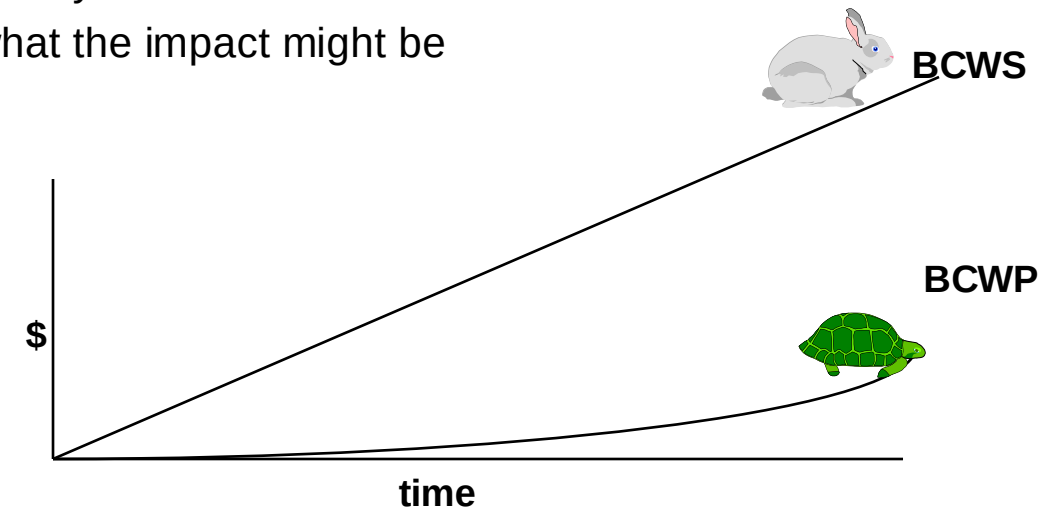
BUDGET: $10,000

	TIER I MILESTONE			⇧ KT AWD					⇧ KT COMP	
WP#	WORK DESCRIPTION	EV METHOD		MONTH 1	MONTH 2	MONTH 3	MONTH 4	MONTH 5	MONTH 6	TOTAL BAC
1	Procure Anvil	0/100	**BCWS**					1,500		1,500
			BCWP					1,500		
2	Paint Fake Tunnel	50/50	**BCWS**			500	500			1,000
			BCWP			500		500		
3	Build Road	units complete	**BCWS**	600	600	600	600	600		3,000
			BCWP	600	600	-	1,200	600		
4	Build ASM	milestone	**BCWS**		1,000	1,000	1,000			3,000
					1	2	3			
			BCWP		1,000	-	1,000	1,000		
5	Install ASM	% complete	**BCWS**				500	500	500	1,500
			BCWP				-	300	1,200	
TOTAL CONTROL ACCOUNT PLAN			**BCWS**	600	1,600	2,100	2,600	2,600	500	10,000
			BCWP	600	1,600	500	2,200	3,900	1,200	10,000
	Schedule Variance	month								
		cumulative								
	Actual Costs									
	Cost Variance	month								
		cumulative								

Control

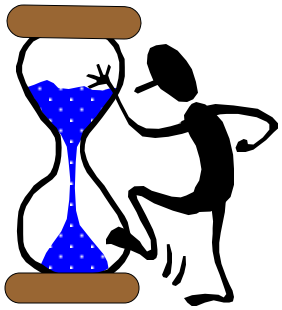
Control

- So, your project has been baselined and work has started
- Is everything going according to plan?
- Next step in the process:
  - figure out your status
  - figure out the problems
  - figure out what you need to do to fix them
  - figure out what the impact might be



Status Reporting

- **Assumption**: An accurate management control system yields accurate data
- **Basic principles**:
  - Report on periodic basic
    - weekly
    - monthly
  - Only ask for the data that you really need and use
    - can eliminate certain formats
      - WBS versus organizational reporting
    - tailor level of reporting to match risk
  - Tailor the data to match how you're managing
    - IPTs?
  - Make it as real time as possible



Contractor Reports

- **Cost Performance Report (CPR)**
  - Format 1: cost and schedule progress by **WBS**
    (specified reporting level usually at level 3)
  - Format 2: cost and schedule progress by **organization**
  - Format 3: changes to performance measurement baseline
  - Format 4: manpower forecast
  - Format 5: variance analysis

current and cum data

Formats 1 and 2

- Contents
  - header:
    - basic contract information (target, ceiling, name of contractor, etc.)
    - range of final estimates
  - body
    - performance data
    - variances
    - budget at completion, estimate at completion

	CURRENT					CUMULATIVE					AT COMPLETION		
	BCWS	BCWP	ACWP	SV	CV	BCWS	BCWP	ACWP	SV	CV	BUDGET	ESTIMATE	VARIANCE
ELEMENT													
ELEMENT													
ELEMENT													
ELEMENT													
ELEMENT													
ELEMENT													
TOTAL													
UB													
PMB													
MR													
TOTAL													

WBS or ORGANIZATION

Cost/Schedule Status Report (C/SSR)

- Similarities to CPR
  - Format 1 (WBS)
  - Format 5 (Problem Analysis)
- Differences
  - Does not require use of approved management system & criteria
  - No current period reporting
  - BCWS and BCWP may be calculated by logical means at higher levels
- Application
  - non-major contracts

Analysis Techniques

or

figuring out where the problems are

Analysis Techniques

- Sort on significant variances
  - eliminate almost complete, just starting, etc.
- Graph and analyze trends
- Look at comparative data
  - e.g. cumulative performance vs. projected performance
- Examine written analysis by contractor
  - does it answer why?
  - adequacy of corrective action plans
- Analysis of schedule trends, critical path
- Analysis of EAC realism

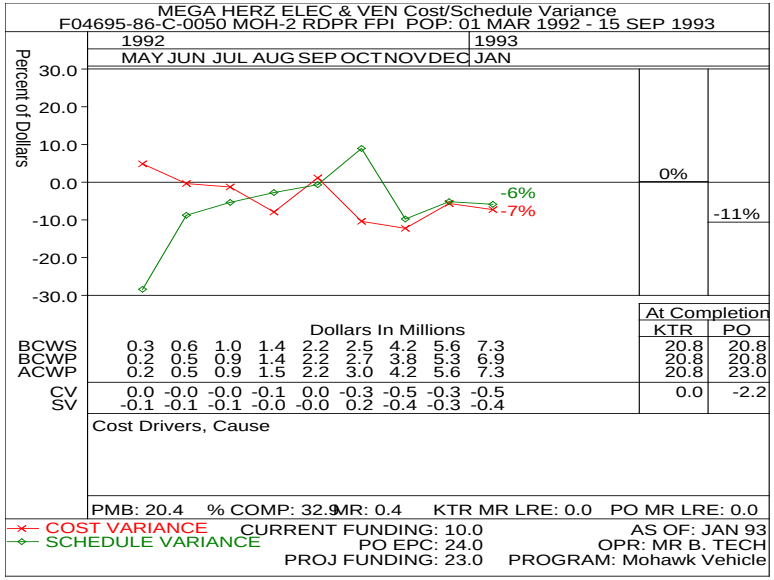
what are the drivers?
what can we do about them?

Where are the significant problems?

sorted by CV $

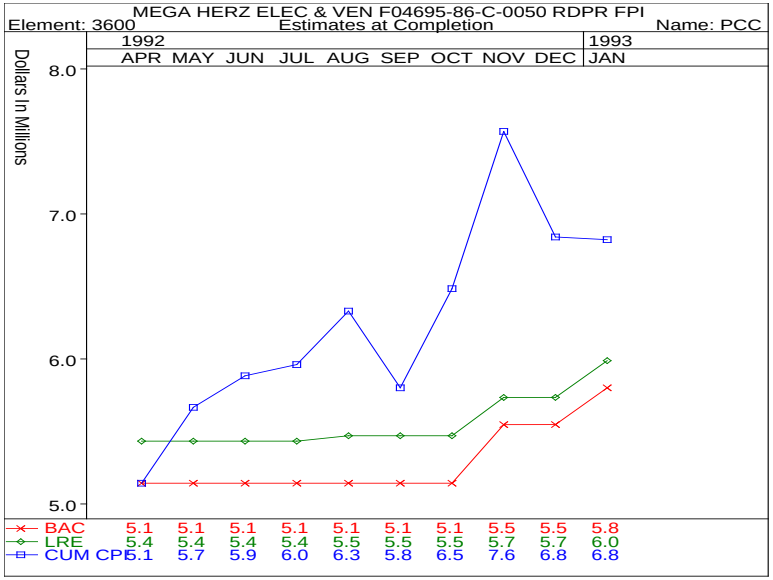
	WBS	DESCRIPTION	Proj Ofcr	%Comp	%Spent	CPI	CV	CV	CV %	VAC	VAC
1	3600	PCC	Zepka	28.99	34.09	0.850	↑	-296.2	-17.62	↔	-187.2
2	3200	COMMUNICATIONS	Tideman	34.63	41.03	0.844	↓	-130.8	-18.49	↔	-87.0
3	G&A	GEN & ADMIN		33.67	36.11	0.932	↓	-45.2	-7.26	↔	-36.8
4	2200	SYS ENGINEERING	Price	85.04	94.35	0.901	↓	-26.4	-10.95	↔	0.0
5	3800	I & A	Troop	35.40	37.08	0.955	↓	-24.2	-4.75	↔	-24.8
6	2100	PROJ MANAGEMENT	Brown	45.70	48.51	0.942	↔	-17.4	-6.16	↔	-3.2
7	2300	FUNC INTEGRA	Price	71.62	75.23	0.952	↓	-17.4	-5.03	↔	-30.8
8	5200	MANAGEMENT DATA	Simmons	84.18	98.10	0.858	↓	-13.2	-16.54	↑	-16.0
9	3100	SENSORS	Smith	20.87	21.49	0.971	↓	-10.6	-2.94	↔	-21.6
10	4000	SPARES	Blair	17.87	18.90	0.945	↑	-7.8	-5.78	↔	-6.2
11	6200	SYSTEM TEST	Hall	60.82	61.66	0.986	↑	-5.6	-1.38	↔	-2.0
12	5100	ENG DATA	Novak	38.51	52.80	0.729	↓	-4.6	-37.10	↔	0.0
13	MR	MGT RESERVE		0.00	0.00			0.0		↔	439.2
14	UB	UNDIST BUDGET						0.0			0.0
15	COM	COST OF MONEY						0.0			0.0
16	3700	DATA DISPLAY	Troop	41.13	41.13	1.000	↔	0.0	0.00	↔	0.0
17	OV	OVERHEAD						0.0			0.0
18	6100	TEST FACILITIES	Smart	100.00	98.02	1.020	↔	2.0	1.98	↔	0.0
19	3500	COMP PROGRAMS	Pino	46.46	44.66	1.040	↓	3.4	3.87	↔	-1.4
20	6300	PCC TEST	Bond	23.13	22.64	1.021	↓	4.2	2.10	↔	0.0
21	3400	ADPE	Zepka	41.89	39.79	1.053	↓	12.6	5.02	↔	4.6
22	3300	AUX EQUIP	Tideman	27.57	24.33	1.133	↓	78.2	11.73	↓	8.4

Graphing Techniques



Overall cost and schedule trend

EAC realism



graphs show overall trend... are you getting better, or worse?

Analysis of Variances

CURRENT OR CUM TO DATE

Schedule Variance

SV ($) = BCWP - BCWS

$$SV\ (\%) = \frac{BCWP - BCWS}{BCWS} \times 100\%$$

Cost Variance

CV ($) = BCWP - ACWP

$$CV\ (\%) = \frac{BCWP - ACWP}{BCWP} \times 100\%$$

CASE STUDY

Part 4

Case Study - Accounting Data

• Month 1	road	$700	700
• Month 2	road	$720	
	drawings	$980	1,700
• Month 3	road	$300	
	tunnel	$200	1,300
	ASM	$800	
• Month 4	road	$1,500	
	tunnel	$400	2,300
	ASM	$400	
• Month 5	road	$700	
	tunnel	$500	
	ASM	$1,900	5,200
	anvil	$1,700	
	install	$400	
• Month 6	install	$2,100	2,100
• total			$13,300

							EV Techniques	0/100, 50/50, Units Complete, % Complete, Milestones	

CONTROL ACCT. TITLE: Roadrunner CONTROL ACCOUNT MANAGER: Wile E. Coyote

BUDGET: $10,000

	TIER I MILESTONE			⇧ KT AWD					⇧ KT COMP	
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			BCWP					1,500		
2	Paint Fake Tunnel	50/50	**BCWS**			500	500			1,000
			BCWP			500		500		
3	Build Road	units complete	**BCWS**	600	600	600	600	600		3,000
			BCWP	600	600	-	1,200	600		
4	Build ASM	milestone	**BCWS**		1,000	1,000	1,000			3,000
					1	2	3			
			BCWP		1,000	-	1,000	1,000		
5	Install ASM	% complete	**BCWS**				500	500	500	1,500
			BCWP				-	300	1,200	
TOTAL CONTROL ACCOUNT PLAN			**BCWS**	600	1,600	2,100	2,600	2,600	500	10,000
			BCWP	600	1,600	500	2,200	3,900	1,200	10,000
	Schedule Variance	month								
		cumulative								
	Actual Costs									
	Cost Variance	month								
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EV Techniques	0/100, 50/50, Units Complete, % Complete, Milestones

CONTROL ACCT. TITLE: Roadrunner

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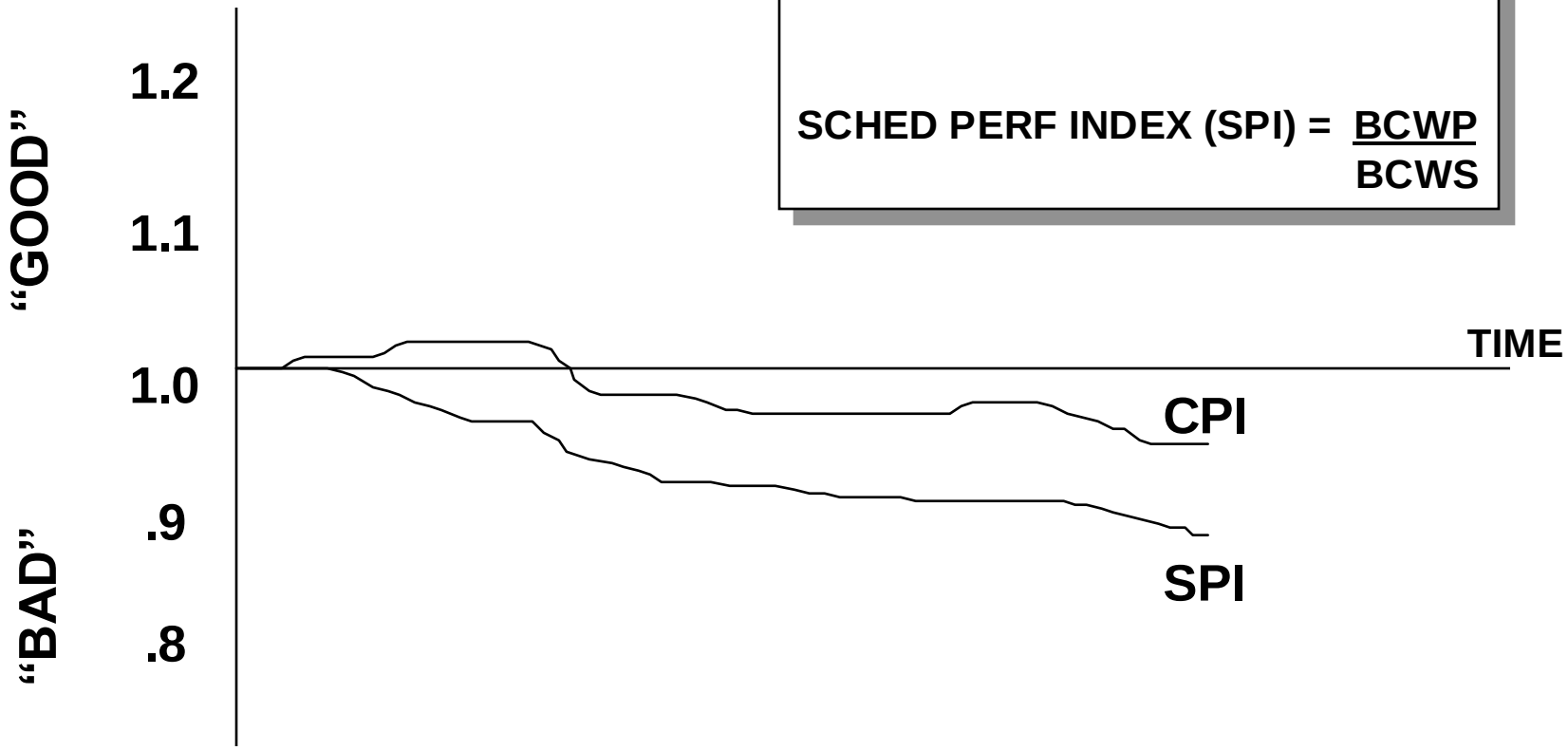
			BCWP			500		500		
3	Build Road	units complete	**BCWS**	600	600	600	600	600		3,000
			BCWP	600	600	-	1,200	600		
4	Build ASM	milestone	**BCWS**		1,000	1,000	1,000			3,000
					1	2	3			
			BCWP		1,000	-	1,000	1,000		
5	Install ASM	% complete	**BCWS**				500	500	500	1,500
			BCWP				-	300	1,200	
TOTAL CONTROL ACCOUNT PLAN			**BCWS**	600	1,600	2,100	2,600	2,600	500	10,000
			BCWP	600	1,600	500	2,200	3,900	1,200	10,000
	Schedule Variance	month		0	0	-1,600	-400	1,300	700	
		cumulative		0	0	-1,600	-2,000	-700	0	
	Actual Costs			700	1,700	1,300	2,300	5,200	2,100	13,300
	Cost Variance	month		-100	-100	-800	-100	-1,300	-900	
		cumulative		-100	-200	-1,000	-1,100	-2,400	-3,300	

Performance Indices

$$\text{COST PERF INDEX (CPI)} = \frac{\text{BCWP}}{\text{ACWP}}$$

$$\text{SCHED PERF INDEX (SPI)} = \frac{\text{BCWP}}{\text{BCWS}}$$

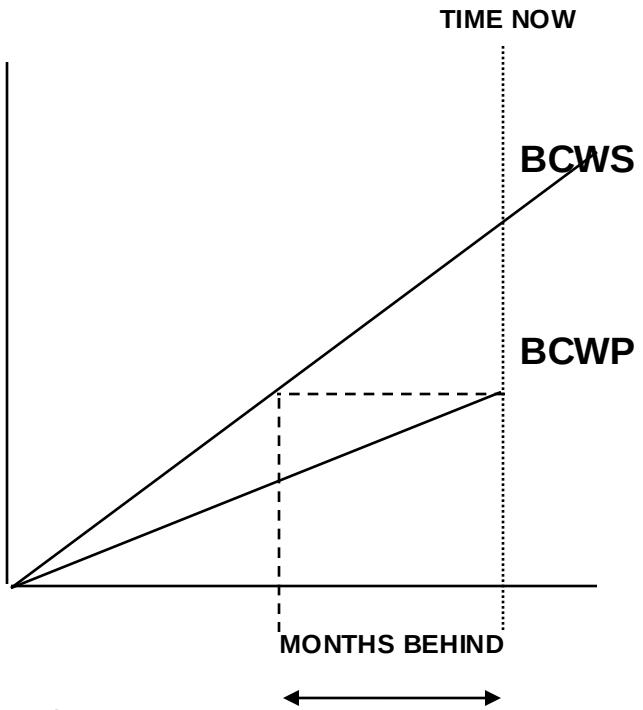


Schedule Status

$$\% \text{ scheduled} = \frac{\text{BCWS}}{\text{BAC}} \times 100\%$$

compare

$$\% \text{ completed} = \frac{\text{BCWP}}{\text{BAC}} \times 100\%$$

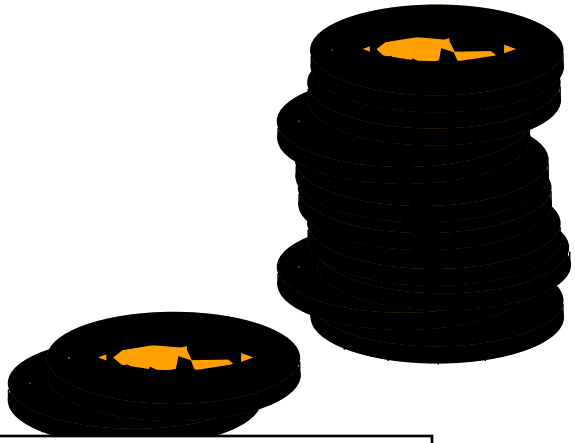


$$\text{Months ahead or behind} = \frac{\text{SV \$}}{\text{Average monthly BCWS \$}}$$

Budget Status

budget status

$$\% \text{ spent} = \frac{ACWP}{BAC} \times 100\%$$



compare:

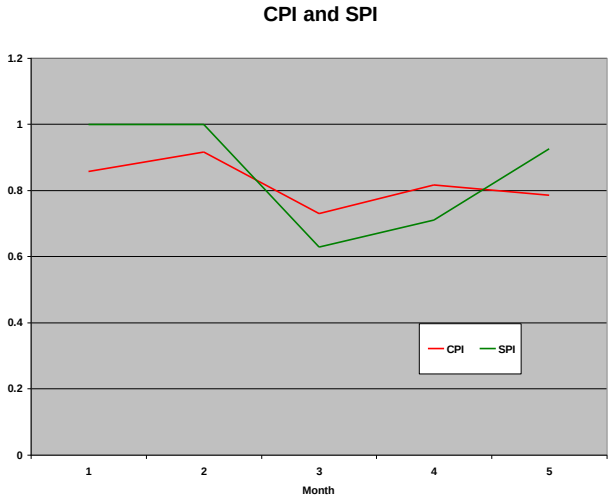
% spent vs. % complete

example: 60% spent vs. 50% complete

Significant Variances

- What is a **<u>significant</u>** variance?
  - % variance (e.g., >10%)
  - $ variance (e.g., >$50,000)
  - critical path element
  - risk/complexity
  - impact to other elements
  - Top 10, Top 20, etc.
  - contractor defined

Case Study Analysis
(month 5)



Variance Analysis Report

The program is now 88% complete. We have now spent $1,200 more than our original budget, primarily driven by two problems:

1) building the sections of roads (unforeseen grading problems)

2) manufacturing and quality testing of the ASM (increase in the price of raw stock and higher rates than forecast for quality personnel)

We are only 20% complete on installation, instead of our scheduled 66%. In order to meet contract schedule, we will have to expend overtime. Costs will increase by an expected $1,000.

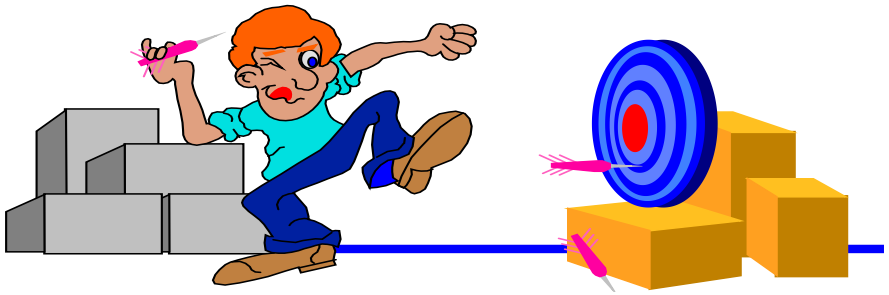
Performance Report, Month 5

	BCWS	BCWP	ACWP	SV	CV	CV %
Procure Anvil	1,500	1,500	1,700	0	-200	-13%
Paint Fake Tunnel	1,000	1,000	1,100	0	-100	-10%
Build Road	3,000	3,000	3,920	0	-920	-31%
Build ASM	3,000	3,000	4,080	0	-1,080	-36%
Install ASM	1,000	300	400	-700	-100	-33%
	9,500	8,800	11,200	-700	-2,400	

	BCWS	BCWP	ACWP	SV	CV	CV %
Build ASM	3,000	3,000	4,080	0	-1,080	-36%
Build Road	3,000	3,000	3,920	0	-920	-31%
Procure Anvil	1,500	1,500	1,700	0	-200	-13%
Paint Fake Tunnel	1,000	1,000	1,100	0	-100	-10%
Install ASM	1,000	300	400	-700	-100	-33%

What will be the final cost?

- **<u>Estimate at Completion (EAC)</u>**
  - defined as actual cost to date + estimated cost of work remaining
  - contractor develops comprehensive EAC at least annually
    - reported by WBS in cost performance report
  - should examine on monthly basis
  - consider the following in EAC generation
    - performance to date
    - impact of approved corrective action plans
    - known/anticipated downstream problems
    - best estimate of the cost to complete remaining work
  - also called latest revised estimate (LRE), indicated final cost, etc.



ACWP + ETC = EAC

One method: statistical formulae

- **Common EAC Formulae:**

$$EAC = \frac{BAC}{CPI}$$

$$= ACWP_{cum} + \frac{\text{Budgeted Cost of Work Remaining}}{CPI_3}$$

$$= ACWP_{cum} + \frac{\text{Budgeted Cost of Work Remaining}}{.8(CPI) + .2(SPI)}$$

$$= ACWP_{cum} + \frac{\text{Budgeted Cost of Work Remaining}}{CPI * SPI}$$

Other methods of EAC calculation

- “Grass Roots” or formal EAC
  - detailed build-up from the lowest level detail
  - hours, rates, bill of material, etc.
- Average of statistical formulae
- Show range of EACs (optimistic, most probable, pessimistic)
- Complete schedule risk analysis for remaining work, estimate work remaining

Why do we need accurate EACs?

- **Variance at Completion vs. Contractor Loss**
  - **Positive VAC:**
    - EAC < BAC underrun contractor gain
  - **Negative VAC**:
    - EAC > BAC share area contractor partial loss
    - EAC > ceiling overrun contractor loss (100%)
- **Customer develops top level EAC for comparison**
  - customer will limit progress payments if EAC is greater than ceiling
  - customer needs forecast of fund requirements
- **May still have time to change the final outcome**

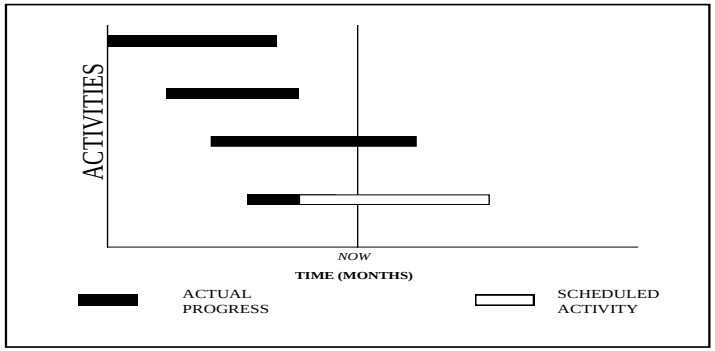
Additional References

SCHEDULE TYPES

- **Gantt Chart**
  - Activities Show a Specific Start and Stop Date
- **Milestone Chart**
  - Major Event Oriented. Shows Start or Stop Date of Activity
- **Line of Balance**
  - Depicts Production Activity. Actual versus Planned Output
- **Networking**
  - Identifies and Defines all Activities and Events for a Program and Links Them in Logical "Cause and Effect" Sequences

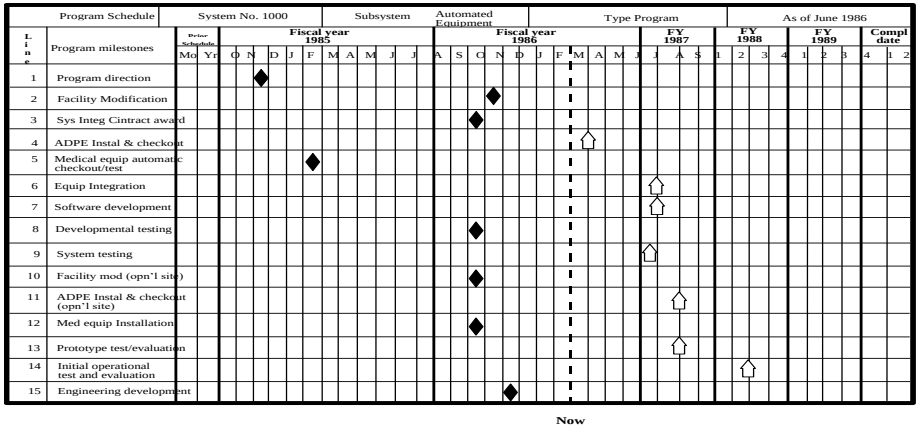
SCHEDULE TYPES (cont'd)

GANTT CHART



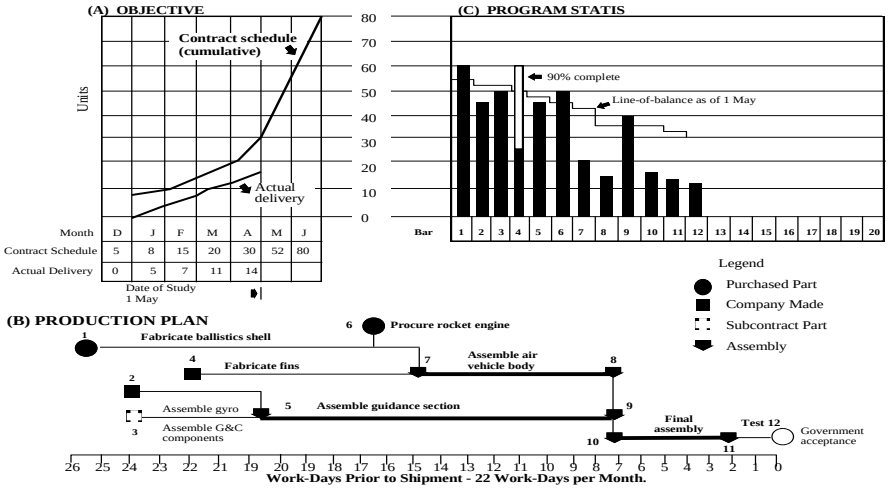
Attachment 3

MILESTONE CHART

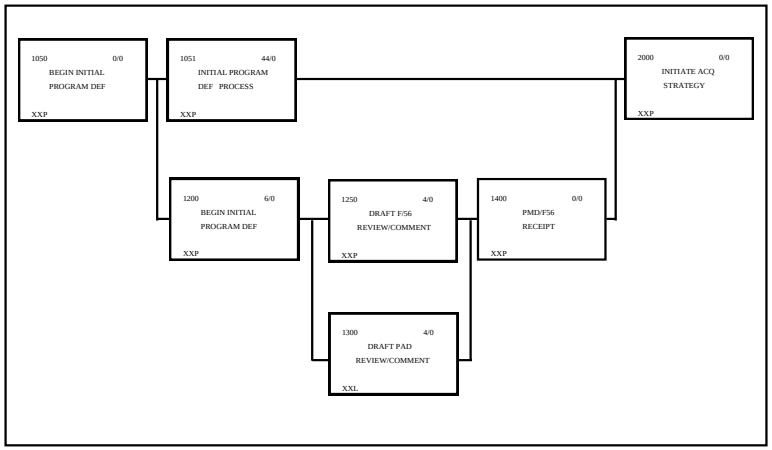


Attachment 6

LINE OF BALANCE TECHNIQUE



NETWORK



Attachment 1

Key Documents in Understanding EVMS

- Gary Christle, 22 Jan 94, paper, Implementation of Earned Value - A Model Program Approach
- USD(A&T) Letter, 25 Jan 94, Improved Cost and Schedule Performance Management
- Gary Christle, 15 Sep 94, paper, The Cost/Schedule Control Systems Criteria and Earned Value Management: A Vision
- USD(A&T) Letter, 1 Oct 96, Compliance Responsibility for the Cost/Schedule Control Systems Criteria (C/SCSC)
- 3 Oct 97, Earned Value Management Implementation Guide (Rev 1)
- USD(A&T) Letter, 14 Dec 96, Industry Standard "Guidelines for Earned Value Management Systems"
- Wayne Abba, article in Jan-Feb 97 Program Manager magazine, Earned Value Management - Reconciling Government and Commercial Practices
- ANSI/EIA 748-1998, Earned Value Management Systems
- Quentin W. Fleming & Joel M. Koppelman, book, Earned Value Project Management
- Earned Value web page: www.acq.osd.mil/pm